



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN022 – Enabling Mathematics B

Differentiations Exam: Sample

This exam represents 35% of the total marks for the subject

Time allowed:	1 hour and 20 minutes
Items permitted in exam:	UOW approved calculator
Aids supplied for exam:	Formula Sheet

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only. Pencils are permitted for diagrams and graphs.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. You must achieve a minimum of mark 40% in the Differentiation Exam. If this requirement is not met, your final mark for the subject will be a Technical Fail
6. Total number of questions: 18
 - Questions 1-6 are multiple choice
 - Each question is worth 1 mark
 - Marks are awarded for the answer only.
 - Circle the letter (A, B, C, D, or E) corresponding to the correct answer of each question.
 - Questions 7-18 are long answer questions
 - The value of each question is indicated.
 - Part marks may be awarded.

No examination material may be removed from the examination venue.

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Question 1

Evaluate the limit, if it exist: $\lim_{x \rightarrow -2} \frac{(x+2)^3}{x^2 + 2x + 5}$

$$\lim_{x \rightarrow -2} \frac{(x+2)^3}{x^2 + 2x + 5} = \frac{0}{5} = 0$$

OPTION A**Question 2**

Given $f(x) = x^2 - 6x - 1$ find the value of x so that $f'(x) = 0$

$$f(x) = x^2 - 6x - 1$$

$$f'(x) = 2x - 6$$

$$f'(x) = 0 \Rightarrow 2x - 6 = 0 \Rightarrow x = 3$$

OPTION D**Question 3**

Find the gradient function of the curve $\frac{1}{\sqrt[3]{x^2 - 1}}$.

$$y = \frac{1}{\sqrt{x^2 - 1}} = (x^2 - 1)^{-1/3}$$

$$y' = \frac{-1}{3} (x^2 - 1)^{-4/3} \times 2x$$

$$y' = \frac{-2x}{3} (x^2 - 1)^{-4/3}$$

$$y' = -\frac{2x}{(x^2 - 1)^{4/3}} = -\frac{2x}{\sqrt[3]{(x^2 - 1)^4}}$$

OPTION C**Question 4**

Find the derivative of $f(x) = \frac{3x^2 + 1}{x^3}$

$$f(x) = \frac{3x^2 + 1}{x^3} = \frac{3}{x} + \frac{1}{x^3} = 3x^{-1} + x^{-3}$$

$$f'(x) = -3x^{-2} - 3x^{-4}$$

$$= -\frac{3}{x^2} - \frac{3}{x^4}$$

OPTION E**Question 5**

Find the derivative of $y = \frac{\sin(3x)}{e^x}$.

$$y = \frac{\sin(3x)}{e^x}$$

quotient rule

$$u = \sin 3x \quad v = e^x$$

$$u' = 3 \cos 3x \quad v' = e^x$$

$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$

$$= \frac{3 \cos 3x e^x - \sin 3x e^x}{e^{2x}} = \frac{3e^x \cos 3x - e^x \sin 3x}{e^{2x}}$$

OPTION A

Question 6

Find the equation of the tangent to the curve $y = 3x^2$ at $x = -1$. Write your answer in gradient-intercept form.

$$y = 3x^2$$

$$y' = 6x$$

$$y'(-1) = -6$$

When $x = -1$,

$$y = 3(-1)^2$$

$$= 3$$

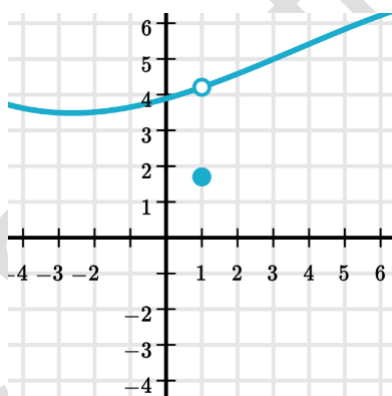
$$y - y_1 = m(x - x_1) \rightarrow y - 3 = -6(x + 1)$$

$$y - 3 = -6x - 6 \rightarrow y = -6x - 3$$

OPTION A

Question 7 (1 marks)

Consider the graph drawn below. Determine whether $\lim_{x \rightarrow 1} f(x)$ exists. Justify your answer.



$$\lim_{x \rightarrow 1} f(x) = 4$$

As the curve approaches 4 from the left and right side of $x = 1$.

Question 8 (4 marks)

Evaluate the following, if they exist:

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4} &= \lim_{x \rightarrow 2} \frac{(x+1)(x-2)}{(x-2)(x+2)} \\ &= \lim_{x \rightarrow 2} \frac{(x+1)}{(x+2)} \\ &= \frac{2+1}{2+2} \\ &= \frac{3}{4} \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \lim_{x \rightarrow \infty} \frac{6x^3 - x + 2}{3x^2 + 7} \\
 \lim_{x \rightarrow \infty} \frac{6x^3 - x + 2}{3x^2 + 7} = \frac{\infty}{\infty} \text{ (indeterminate form)} \\
 \therefore \lim_{x \rightarrow \infty} \frac{6x^3 - x + 2}{3x^2 + 7} = \frac{\frac{6x^3}{x^3} - \frac{x}{x^3} + \frac{2}{x^3}}{\frac{3x^2}{x^3} + \frac{7}{x^3}} \\
 = \lim_{x \rightarrow \infty} \frac{6 - \frac{1}{x^2} + \frac{2}{x^3}}{\frac{3}{x} + \frac{7}{x^3}} \\
 = \frac{6}{0} = DNE
 \end{aligned}$$

Question 9 (2 marks)

Find $\frac{dy}{dx}$ of $y = 2x + \frac{\sqrt{x}}{x}$, leaving your answer in surd form.

$$\begin{aligned}
 f(x) &= 2x + \frac{\sqrt{x}}{x} = 2x + x^{1/2-1} = 2x + x^{-1/2} \\
 f'(x) &= 2 + \frac{-1}{2} x^{-3/2} \\
 &= 2 + \frac{-1}{2x^{3/2}} \\
 &= 2 - \frac{1}{2\sqrt{x^3}}
 \end{aligned}$$

Question 10 (3 marks)

Find the gradient of $(x^2 - 1)(2x + 3)^4$ and simplify your answer fully.

$$y = (x^2 - 1)(2x + 3)^4$$

product rule

$$u = x^2 - 1 \quad v = (2x + 3)^4$$

$$u' = 2x \quad v' = 8(2x + 3)^3$$

$$\frac{dy}{dx} = u'v + uv'$$

$$\begin{aligned}
 &= 2x(2x + 3)^4 + (x^2 - 1)8(2x + 3)^3 \\
 &= 2(2x + 3)^3 (x(2x + 3) + 4(x^2 - 1)) \\
 &= 2(2x + 3)^3 (2x^2 + 3x + 4x^2 - 4) \\
 &= 2(2x + 3)^3 (6x^2 + 3x - 4)
 \end{aligned}$$

Question 11 (3 marks)

Find the second derivative of the function $y = \sqrt{x}(2 - \sqrt{x})$, writing your answer in surd form.

$$\begin{aligned}
 y &= \sqrt{x}(2 - \sqrt{x}) \\
 &= x^{\frac{1}{2}} \left(2 - x^{\frac{1}{2}} \right) \\
 &= 2x^{\frac{1}{2}} - x \\
 \frac{dy}{dx} &= 2 \times \frac{1}{2} x^{-\frac{1}{2}} - 1 = x^{-\frac{1}{2}} - 1 \\
 \frac{d^2y}{dx^2} &= -\frac{1}{2} x^{-\frac{3}{2}} \\
 &= -\frac{1}{2x^{\frac{3}{2}}} = -\frac{1}{2\sqrt{x^3}}
 \end{aligned}$$

Question 12 (3 marks)

Find $f'(1)$ for $f(x) = (e^{-x^2+2})^3$ by using the chain rule. Leave your answer in exact form.

$$\begin{aligned}
 f(x) &= (e^{-x^2+2})^3 \\
 &\quad \text{chain rule} \\
 f'(x) &= 3(e^{-x^2+2})^2 (-2xe^{-x^2+2}) \\
 &= -6xe^{-x^2+2} (e^{-x^2+2})^2 \left[= -6x(e^{-x^2+2})^3 \right] \\
 f'(1) &= -6 \times 1 e^{-(1)^2+2} (e^{-(1)^2+2})^2 \\
 &= -6e \times e^2 = -6e^3
 \end{aligned}$$

Question 13 (3 marks)

Differentiate the functions $y = \frac{\ln 2x}{x^2}$ and simplify your answer fully if possible.

$$\begin{aligned}
 y &= \frac{\ln 2x}{x^2} \\
 &\quad \text{quotient rule} \\
 u &= \ln 2x \quad v = x^2 \\
 u' &= \frac{2}{2x} = \frac{1}{x} \quad v' = 2x \\
 \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\
 &= \frac{\frac{1}{x} \times x^2 - \ln 2x \times 2x}{(x^2)^2} = \frac{x - 2x \ln 2x}{x^4} \\
 &= \frac{x(1 - 2 \ln 2x)}{x^4} = \frac{1 - 2 \ln 2x}{x^3}
 \end{aligned}$$

Question 14 (3 marks)

Evaluate $f'(2)$ if $f(x) = \ln \frac{(2x-3)}{(x-1)}$.

$$\begin{aligned}f(x) &= \ln \frac{(2x-3)}{(x-1)} \\&= \ln(2x-3) - \ln(x-1) \\f'(x) &= \frac{2}{2x-3} - \frac{1}{x-1} \\f'(2) &= \frac{2}{2 \times 2 - 3} - \frac{1}{2-1} \\&= 2 - 1 = 1\end{aligned}$$

Evaluate $f'(2)$ if $f(x) = \ln(2x-3)(x-1)$.

$$\begin{aligned}f(x) &= \ln(2x-3)(x-1) \\&= \ln(2x-3) + \ln(x-1) \\f'(x) &= \frac{2}{2x-3} + \frac{1}{x-1} \\f'(2) &= \frac{2}{2 \times 2 - 3} + \frac{1}{2-1} \\&= 2 + 1 = 3\end{aligned}$$

Evaluate $f'(2)$ if $f(x) = \ln \sqrt{(2x-3)}$.

$$\begin{aligned}f(x) &= \ln \sqrt{(2x-3)} \\&= \ln(2x-3)^{\frac{1}{2}} \\f'(x) &= \frac{1}{2} \ln(2x-3)^{-\frac{1}{2}} \times \frac{2}{2x-3} \\f'(x) &= \frac{1}{2x-3} \ln(2x-3)^{-\frac{1}{2}} \\&= \frac{1}{(2x-3) \ln \sqrt{(2x-3)}} \\f'(2) &= \frac{1}{(2 \times 2 - 3) \ln \sqrt{(2 \times 2 - 3)}} \\&= \frac{1}{\sqrt{0}} = dne\end{aligned}$$

Question 15 (3 marks)

For the function $f(x) = 2\pi \tan(2x)$, evaluate $f'\left(\frac{\pi}{6}\right)$. Give your answer as an exact value.

$$\begin{aligned}
 f(x) &= 2\pi \tan(2x) \\
 f'(x) &= 2\pi \sec^2(2x) \times 2 \\
 &= 4\pi \sec^2(2x) \\
 f'\left(\frac{\pi}{6}\right) &= 4\pi \sec^2\left(2 \times \frac{\pi}{6}\right) \\
 &= 4\pi \sec^2\left(\frac{\pi}{3}\right) \\
 &= 4\pi \times \frac{1}{\left(\cos\left(\frac{\pi}{3}\right)\right)^2} \\
 &= 4\pi \times \frac{1}{\left(\frac{1}{2}\right)^2} = 4\pi \times 4 = 16\pi
 \end{aligned}$$

Question 16 (3 marks)

Find the derivative of $f(x) = \frac{e^{2x}}{\tan 2x}$ and simplify your answer if necessary.

$$\begin{aligned}
 f(x) &= \frac{e^{2x}}{\tan 2x} \\
 &\text{quotient rule} \\
 u &= e^{2x} \quad v = \tan 2x \\
 u' &= 2e^{2x} \quad v' = 2\sec^2 x \\
 \frac{dy}{dx} &= \frac{2e^{2x} \tan 2x - e^{2x} 2\sec^2 x}{\tan^2 2x} \\
 &= \frac{2e^{2x} \tan 2x - 2e^{2x} \sec^2 x}{\tan^2 2x} \\
 &= \frac{2e^{2x} (\tan 2x - \sec^2 x)}{\tan^2 2x}
 \end{aligned}$$

Question 17 (3 marks)

Find the equation of the tangent to the curve $y = 2e^{\frac{x}{2}} + 1$ at the point where $x = 0$. Write the equation in gradient intercept form.

$$\begin{aligned}
 y &= 2e^{\frac{x}{2}} + 1 \\
 \frac{dy}{dx} &= 2 \times \frac{1}{2} e^{\frac{x}{2}} = e^{\frac{x}{2}}
 \end{aligned}$$

To find m sub in $x = 0$ into $\frac{dy}{dx}$

$$m = e^{\frac{0}{2}} = e^0 = 1$$

To find y_1 sub $x = 0$ into y

$$y = 2e^{\frac{0}{2}} + 1 = 2e^0 + 1 = 2 + 1 = 3$$

Equation of the curve using $y - y_1 = m(x - x_1)$

$$y - 3 = 1(x - 0)$$

$$y = x + 3$$

Question 18 (4 marks)

Find the coordinates of the point on the curve $y = x^2 + 9x + 4$ where the tangent is parallel to the line $3x + y - 1 = 0$.

$$y = x^2 + 9x + 4$$

$$y' = 2x + 9$$

Solve for $y' = m = -3$

$$2x + 9 = -3$$

$$x = -6,$$

When $x = -6$,

$$y = (-6)^2 + 9(-6) + 4$$

$$= -14$$

$$(-6, -14)$$

Or

Find the equation of the tangent to the curve $y = x^2 + 9x + 4$ at the point where the curve crosses the y -axis. Write the equation in general form.

$$y = x^2 + 9x + 4$$

curve crosses the y -axis at $x = 0$

$$y = 0^2 + 9 \times 0 + 4$$

$$= 4$$

$$y' = 2x$$

$$y'(4) = 2 \times 4 = 8$$

The point is $(0, 4)$

$$y' = 2x + 9$$

$$y'(0) = 2 \times 0 + 9 = 9$$

$$y - y_1 = m(x - x_1)$$

$$y - 4 = 9(x - 0)$$

$$9x - y + 4 = 0$$

END OF EXAM